

Abhishek Srivastava

Scientific Machine Learning / Bayesian Inverse Problems / UQ / Computational Fluid Dynamics

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Research Profile

Ph.D. in Mechanical Engineering from IIT Delhi with interdisciplinary research experience spanning computational fluid dynamics, multiphysics modeling, fluid–particle interactions, scientific machine learning, explainable AI, surrogate modeling, optimization, Bayesian inverse problems, and uncertainty quantification. Doctoral research integrated high-fidelity multiphysics simulations with machine learning to study charged-particle dynamics and collection. Current research at IIT Palakkad focuses on deep-learning-based inverse modeling for complex physical systems. Interested in physics-aware and probabilistic machine learning for computational science, engineering inverse problems, and Uncertainty quantification.

Research Expertise

Scientific ML	Surrogate modeling; deep learning; variational autoencoders; physics-aware ML; probabilistic modeling
Bayesian / UQ	Bayesian inverse problems; MCMC; Adaptive Metropolis; HMC; posterior approximation; Fisher Information; covariance analysis
Physics / Simulation	CFD; multiphysics; electrostatics; particle tracing; fluid–particle interactions; computational mechanics
XAI / Interpretation	SHAP; Partial Dependence Plots (PDP); Accumulated Local Effects (ALE); physical interpretation of ML models
Optimization	NSGA-II; genetic algorithms; multi-objective optimization; surrogate-assisted optimization; clustering-guided design
Computing	Python; TensorFlow; Scikit-learn; NumPy; SciPy; Pandas; COMSOL Multiphysics; FEniCS; Git/GitHub

Research Experience

Project Associate, Department of Mechanical Engineering, IIT, Palakkad *Feb. 2026 – Present*
Supervised by Dr. Arijit Hazra

- Working on deep learning-based methodologies for inverse problems and uncertainty quantification in heat transfer and fluid flow problems
- Developing deep-learning-based inverse modeling frameworks for complex physical systems.
- Developing computationally efficient approaches for posterior approximations.
- Comparison of deep learning based posterior estimates with sampling based Bayesian methods.
- Investigating uncertainty quantification and posterior covariance estimation
- Studying information geometry and Fisher Information Matrix-based characterization of inverse problems

Ph.D. Researcher, Department of Mechanical Engineering, IIT Delhi *Jan. 2020 – July 2026*
Supervisors: Dr. Bahni Ray and Dr. Mayank Kumar

- Development of a high-fidelity multiphysics computational framework integrating computational fluid dynamics, electrostatics, and particle tracing to investigate charged aerosol transport and optimize electrostatic particle collection by charge spherical collector.
- Developed an interpretable machine learning framework integrating LightGBM, SHAP, and Partial Dependence Plots (PDP) to analyze charged particle rebound, identify dominant physical mechanisms, and optimize electrostatic particle collection.
- Developed a clustering-guided surrogate-assisted multiobjective optimization framework integrating machine learning, K-means clustering, NSGA-II, TOPSIS, and explainable AI to optimize electrostatic particle collection while minimizing particle rebound.
- Combined simulation, machine learning, explainability, clustering, and optimization for analyzing compact ozone-free electrostatic particle-collection systems.

Research Highlights & Current Direction

Doctoral Research: Surrogate Assisted Explainable Machine Learning Framework for Particle Collection. Developed an end-to-end computational framework integrating multiphysics CFD/electrostatic simulation, particle dynamics, machine-learning surrogate modeling, Explainable AI, clustering, and multi-objective optimization. The framework was used to identify influential physical parameters and optimize charged-particle collection and rebound behavior.

Current Research: Bayesian Inverse Problems and Probabilistic Deep Learning. Developing deep-learning-based approaches for inverse problems, with emphasis on posterior approximation, uncertainty quantification,

probabilistic generative models, and comparison with sampling-based Bayesian inference.

Education

Indian Institute of Technology Delhi Ph.D., Mechanical Engineering; Fluid Mechanics and Thermal Sciences	<i>Jan. 2020 – Feb. 2026</i> GPA: 8.79
Maulana Azad National Institute of Technology, Bhopal M.Tech., Maintenance Engineering	<i>Jul. 2017 – May 2019</i> GPA: 8.16
Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh B.Tech., Mechanical Engineering	<i>Aug. 2011 – Jun. 2015</i> GPA: 7.08

Selected Publications

- [1] **Abhishek Srivastava**, Debabrat Biswal, Bahni Ray, Mayank Kumar, and Debabrata Dasgupta. “Influence of nonuniform electric field due to a spherical collector on the particles’ dynamics in quiescent air.” *Journal of Electrostatics*, 129 (2024), 103916.
- [2] **Abhishek Srivastava**, Sagar Kandpal, Bahni Ray, Mayank Kumar, and Debabrata Dasgupta. “Analyzing the Impact of Charge Particle Rebound on Spherical Collector Efficiency via Explainable Machine Learning.” *Powder Technology*, 469, Part 2 (2025), 121803.
- [3] **Abhishek Srivastava**, Sagar Kandpal, Bahni Ray, Mayank Kumar, and Debabrata Dasgupta. “Uncovering Hidden Trade-Off Patterns in Charged Particle Collection Using K-Means Clustering and Machine Learning Assisted Multiobjective Optimization.” *Under Review: Engineering Applications of Artificial Intelligence*.
- [4] **Abhishek Srivastava**, Arijit Hazra. "Bayesian Inverse Problems via Variational Autoencoders with Jensen-Shannon Wasserstein Regularization" *Under Preparation*.

Selected Conferences

- **Abhishek Srivastava**, et al. “Electrospray based air pollution mitigation: Droplet and particle interaction.” COMPOLL, IIT Delhi, Nov. 2022.
- **Abhishek Srivastava**, et al. “Dynamic Characteristics of Submicron Particle Deposited on the Charged Spherical Collector.” Conference on Fluid Mechanics and Fluid Power, IIT Roorkee, Dec. 2022.
- **Abhishek Srivastava**, Bahni Ray, and Mayank Kumar. “Numerical investigation on deposition process of polydisperse dust particles on charged spherical pendant collector.” International Workshop on Sustainable Energy, Power and Propulsion, IIT Kanpur, Jan. 2024.

Teaching, Awards & Service

Teaching Assistantship: Intermediate Heat Transfer; Advanced Thermodynamics.

Awards: Best Technical Interaction Award, IWSEPP-2024; MHRD Fellowship, Government of India;

Service: Research Scholar Representative, Student Affairs Committee (SAC), IIT Delhi, Jun. 2023 – May 2024.

References

Dr. Bahni Ray Associate Professor, Mechanical Engineering IIT Delhi bray@mech.iitd.ac.in Faculty Profile	Dr. Mayank Kumar Associate Professor, Mechanical Engineering IIT Delhi kmayank@mech.iitd.ac.in Faculty Profile
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